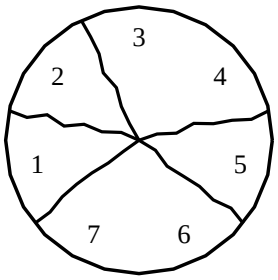
Math Kangaroo Practice

Grade 3-4 · 24 questions · standard · seed 76846

Part A: Each question is worth 3 points

1. A wheel of fortune with the numbers 1 to 7 around its edge is cracked into 5 pieces as shown.

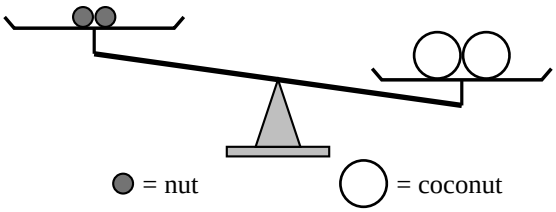


Which piece has the smallest sum of numbers?

A. the piece with the 4 ***B.*** the piece with the 2 ***C.*** the piece with the 5

D. the piece with the 7 ***E.*** the piece with the 1

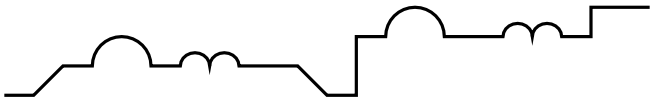
2. On one plate of a balance there are 2 nuts and on the other there are 2 coconuts. When we put one more coconut, exactly like the others, on the plate with the nuts, the balance is level.



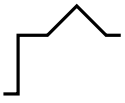
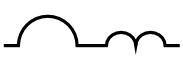
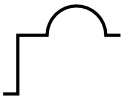
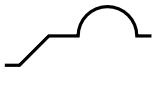
One coconut weighs the same as how many nuts?

A. 5 nuts ***B.*** 4 nuts ***C.*** 3 nuts ***D.*** 2 nuts ***E.*** 1 nut

3. This is the outline of a town seen from the river at sunset.

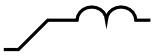
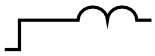


Which piece **cannot** be part of the outline?

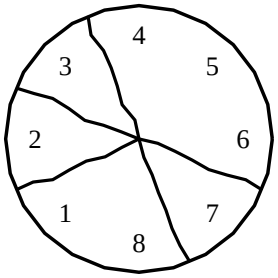
A.  ***B.***  ***C.***  ***D.*** ***E.*** 

4. Far away we see the skyline of a castle.

Which of the pieces **cannot** belong to the skyline?

A. ***B.***  ***C.*** ***D.*** ***E.*** 

5. A round cookie with the numbers 1 to 8 around its edge is cracked into 5 pieces as shown.



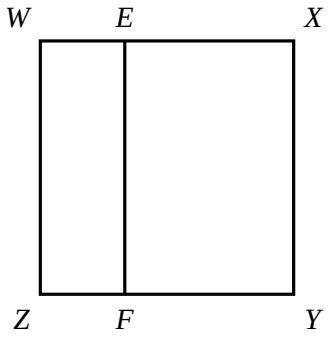
Which piece has the largest sum of numbers?

A. the piece with the 8 ***B.*** the piece with the 2 ***C.*** the piece with the 7
D. the piece with the 6 ***E.*** the piece with the 3

6. Tom, Mia, Julien and Leo each play a different instrument: the trumpet, the harp, the guitar and the piano. Tom blows into the instrument. Leo doesn't play the harp. Leo plays an instrument with strings. Mia plays an instrument with strings.
Who plays the piano?

A. Leo ***B.*** Julien ***C.*** Mia
D. Tom ***E.*** Impossible to determine

7. *WXYZ* is a square. Its side is 6 cm. The line *EF* cuts it into two rectangles, and *WE* = 2 cm.



How many centimetres is the perimeter of the square *WXYZ* larger than that of the rectangle *EXYF*?

A. 20 cm ***B.*** 12 cm ***C.*** 8 cm ***D.*** 4 cm ***E.*** 2 cm

8. Aisha, Omar, Mia and Ravi each play a different instrument: the harp, the trumpet, the violin and the drum. Ravi uses a bow. Omar plays an instrument without strings. Mia blows into the instrument.
Who plays the drum?

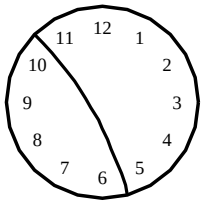
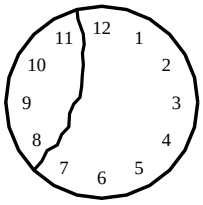
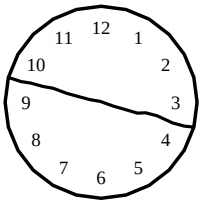
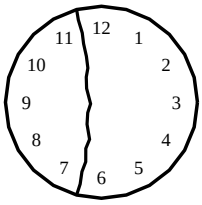
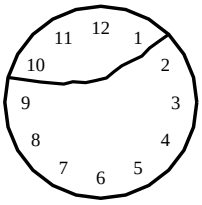
A. Ravi ***B.*** Omar ***C.*** Mia
D. Aisha ***E.*** Impossible to determine

Part B: Each question is worth 4 points

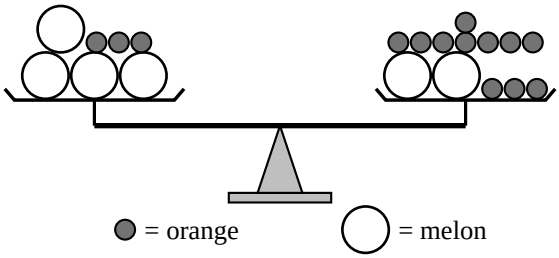
9. Zoe, Julien, Tom and Manon each have a different pet: a goldfish, a cat, a canary and a dog. Manon has a pet that never barks. Tom has a pet that lives in water. Zoe has a pet that can fly. Which statement is true?

A. Manon has a goldfish ***B.*** Tom has a goldfish ***C.*** Tom has a dog
D. Manon has a canary ***E.*** Manon has a dog

10. The face of a clock is cracked into 2 pieces. All the pieces have the same sum of numbers. Which picture shows how it is cracked?

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

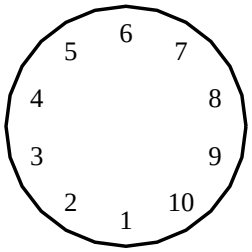
11. The balance is level: 4 melons and 3 oranges on one plate, 2 melons and 11 oranges on the other.



One melon weighs the same as how many oranges?

A. 14 oranges ***B.*** 11 oranges ***C.*** 8 oranges ***D.*** 5 oranges ***E.*** 4 oranges

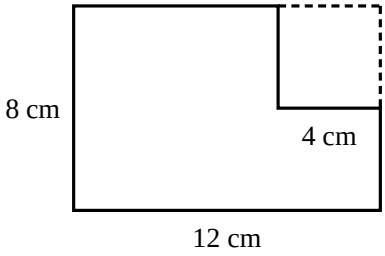
12. A round cookie with the numbers 1 to 10 around its edge is cracked into 2 pieces. The sums of the numbers in the pieces are two consecutive numbers.



What is the smallest of these sums?

A. 27 ***B.*** 28 ***C.*** 29 ***D.*** 33 ***E.*** 34

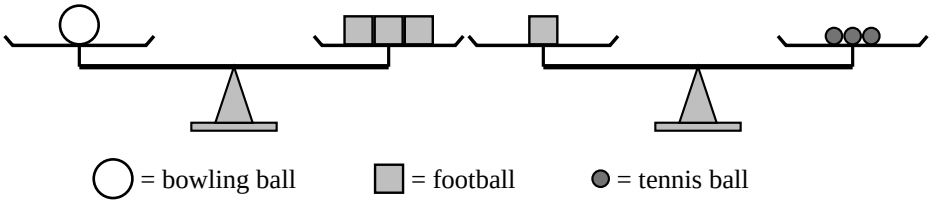
13. A rectangle is 12 cm long and 8 cm wide. A square with side 4 cm is cut off one corner.



What is the perimeter of the remaining shape?

A. 36 cm ***B.*** 40 cm ***C.*** 48 cm ***D.*** 49 cm ***E.*** 56 cm

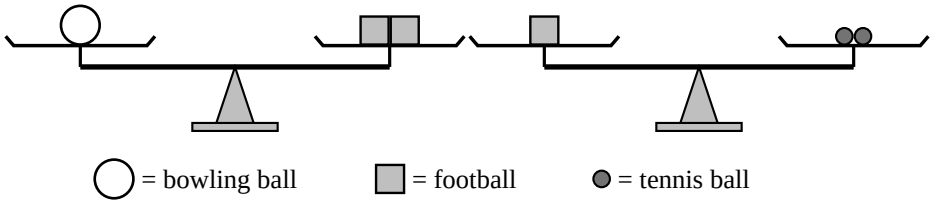
14. The two balances are level. One bowling ball weighs as much as 3 footballs, and one football weighs as much as 3 tennis balls.



How many tennis balls weigh as much as 1 bowling ball?

A. 3 ***B.*** 6 ***C.*** 8 ***D.*** 9 ***E.*** 10

15. The two balances are level. One bowling ball weighs as much as 2 footballs, and one football weighs as much as 2 tennis balls.

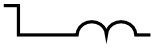


How many tennis balls weigh as much as 1 bowling ball?

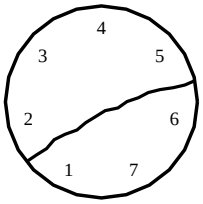
A. 5 ***B.*** 4 ***C.*** 3 ***D.*** 2 ***E.*** 1

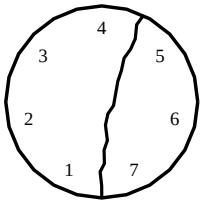
16. This is the outline of a town seen from the river at sunset.

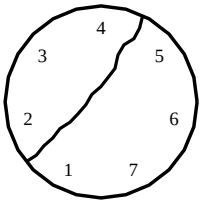
Which piece **can** be part of the outline?

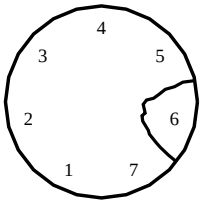
A. ***B.***  ***C.*** ***D.*** ***E.***

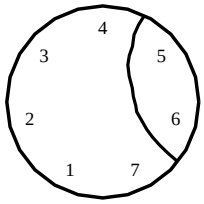
17. A round cake with the numbers 1 to 7 around its edge is cracked into 2 pieces. All the pieces have the same sum of numbers.
Which picture shows how it is cracked?

A. 

B. 

C. 

D. 

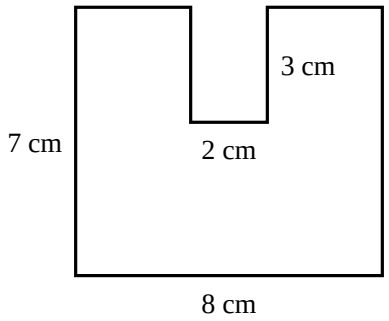
E. 

Part C: Each question is worth 5 points

18. Chloe, Lucas, Tom and Emma each have a different pet: a rabbit, a goldfish, a canary and a turtle. Chloe has a pet that lives in water. Lucas has a pet with a shell. Tom has a bird.
Which statement is **not** true?

A. Chloe has a goldfish ***B.*** Lucas has a turtle ***C.*** Lucas has a goldfish
D. Emma has a rabbit ***E.*** Tom has a canary

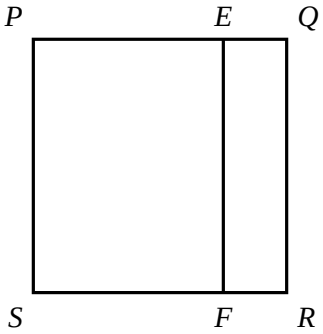
19. A 8 cm by 7 cm rectangle of cardboard has a rectangular notch 2 cm wide and 3 cm deep cut out of the middle of its top side.



What is the perimeter of the new shape?

A. 33 cm ***B.*** 36 cm ***C.*** 40 cm ***D.*** 43 cm ***E.*** 44 cm

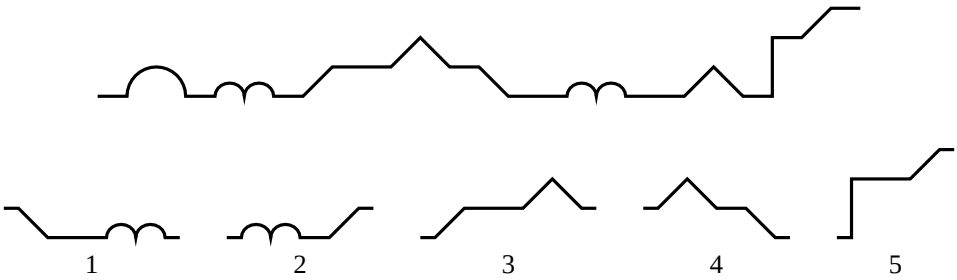
20. The square *PQRS* is cut into two rectangles. The perimeter of *PEFS* is 14 cm and the perimeter of *EQRF* is 10 cm.



What is the perimeter of the square *PQRS*?

A. 16 cm ***B.*** 12 cm ***C.*** 8 cm ***D.*** 6 cm ***E.*** 4 cm

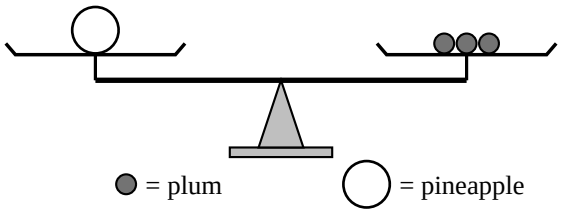
21. Lena cut a strip of paper along this line.



How many of the pieces 1–5 below belong to the line?

A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

22. The balance shows that one pineapple weighs as much as 3 plums. 1 pineapple and 1 plum weigh 600 g altogether.



How much does one plum weigh?

A. 75 g ***B.*** 100 g ***C.*** 125 g ***D.*** 150 g ***E.*** 300 g

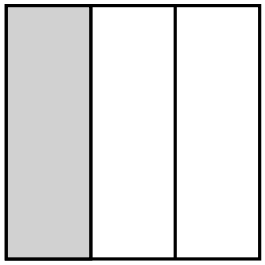
23. Lucas, Ravi, Nora and Nicolas were born in different months: March, July, December and April. Lucas was born in a month that starts with J. Ravi was not born in March. Nora was born in winter.

Which statement is **not** true?

A. Nicolas was born in March ***B.*** Nora was born in December ***C.*** Nora was born in July

D. Lucas was born in July ***E.*** Ravi was born in April

24. Mia cuts a square chocolate bar into 3 equal strips as shown. The perimeter of each strip is 48 cm.



How long is the side of the square?

A. 72 cm ***B.*** 30 cm ***C.*** 24 cm ***D.*** 18 cm ***E.*** 6 cm

Answer Key

1.	**E**	**2.**	**D**	**3.**	**A**	**4.**	**E**	**5.**	**D**	**6.**	**B**
7.	**D**	**8.**	**B**	**9.**	**B**	**10.**	**C**	**11.**	**E**	**12.**	**A**
13.	**B**	**14.**	**D**	**15.**	**B**	**16.**	**E**	**17.**	**A**	**18.**	**C**
19.	**B**	**20.**	**A**	**21.**	**D**	**22.**	**D**	**23.**	**C**	**24.**	**D**

Solutions

1. E (the piece with the 1) — The sums are: 3+4 = 7, 5 = 5, 6+7 = 13, 1 = 1, 2 = 2. [2002 Q11 · biggest_piece]

2. D (2 nuts) — Take 1 coconut off both plates: 2 nuts balance 2 – 1 = 1 coconut. So one coconut = 2 nuts. [2002 Q9 · one_more_balances]

3. A — All the other pieces can be found in the line, but this one cannot: it has a pointed roof where the line has a round dome. [2002 Q7 · cannot_belong]

4. E — All the other pieces can be found in the line, but this one cannot: it has the right shapes but in the wrong order. [2002 Q7 · cannot_belong]

5. D (the piece with the 6) — The sums are: 4+5+6 = 15, 7 = 7, 8+1 = 9, 2 = 2, 3 = 3. [2002 Q11 · biggest_piece]

6. B (Julien) — The only possible matching: Tom – trumpet; Mia – harp; Julien – piano; Leo – guitar. [2002 Q18 · who_has]

7. D (4 cm) — *EXYF* has sides 4 and 6 cm, perimeter 20 cm. Square: 24 cm. Difference 4 cm (= 2 × 2). [2002 Q6 · square_minus_strip]

8. B (Omar) — The only possible matching: Aisha – harp; Omar – drum; Mia – trumpet; Ravi – violin. [2002 Q18 · who_has]

9. B (Tom has a goldfish) — The only possible matching: Zoe – canary; Julien – dog; Tom – goldfish; Manon – cat. [2002 Q18 · which_true]

10. C — All numbers add up to 78, so the pieces must be 39 + 39 = 78. Only this picture gives 39, 39. [2002 Q11 · which_picture]

11. E (4 oranges) — Take 2 melons and 3 oranges off both plates. Left: 2 melons = 8 oranges, so one melon = 8 ÷ 2 = 4. [2002 Q9 · both_sides]

12. A (27) — All the numbers add up to 55. Two consecutive numbers that add up to 55: 27 + 28 = 55. The smallest is 27. [2002 Q11 · largest_sum]

13. B (40 cm) — The two new edges (4 cm and 4 cm) replace the two removed pieces of the outer edge, which were also 4 cm and 4 cm. So the perimeter is the same as the rectangle's: 2 × (12 + 8) = 40 cm. [2002 Q6 · corner_cut]

14. D (9) — One bowling ball = 3 footballs = 3 × 3 = 9 tennis balls. [2002 Q9 · chain]

15. B (4) — One bowling ball = 2 footballs = 2 × 2 = 4 tennis balls. [2002 Q9 · chain]

16. E — Only this piece is found in the line. The others don't fit — A: it is a mirror image of a part of the line; B: it has two small bumps where the line has a round dome; C: it is a mirror image of a part of the line; D: it has two small bumps where the line has a pointed roof. [2002 Q7 · can_belong]

17. A — All numbers add up to 28, so the pieces must be 14 + 14 = 28. Only this picture gives 14, 14. [2002 Q11 · which_picture]

18. C (Lucas has a goldfish) — The only possible matching: Chloe – goldfish; Lucas – turtle; Tom – canary; Emma – rabbit. [2002 Q18 · which_not_true]

19. B (36 cm) — The bottom of the notch (2 cm) replaces the 2 cm taken out of the top edge, and the two sides of the notch add 2 × 3 cm. Perimeter = 2 × (8 + 7) + 2 × 3 = 36 cm. [2002 Q6 · notch]

20. A (16 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 14 + 10 = 24 cm = 6 sides. Side = 24 ÷ 6 = 4 cm, perimeter 4 × 4 = 16 cm. [2002 Q6 · two_perimeters]

21. D (4) — Pieces 1, 2, 3, 4 fit. Not: 5 (its flat part between the two features has the wrong length). [2002 Q7 · how_many_fit]

22. D (150 g) — Swap each pineapple for 3 plums: 1 × 3 + 1 = 4 plums weigh 600 g. One plum: 600 ÷ 4 = 150 g. [2002 Q9 · grams]

23. C (Nora was born in July) — The only possible matching: Lucas – July; Ravi – April; Nora – December; Nicolas – March. [2002 Q18 · which_not_true]

24. D (18 cm) — A strip is the full side long and 1/3 of it wide, so its perimeter is side + side + 2 × (side ÷ 3) = 8 × (side ÷ 3). 48 ÷ 8 = 6, so the side is 3 × 6 = 18 cm. [2002 Q6 · strips_to_square]